

More Yap, Less Meaning:

Uncovering Self-Improvement Behavior in Small Language Models

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Research Question

When a Small Language Model (SLM) gives a wrong answer, can it understand what went wrong and write a useful hint for its second attempt?

Prior work

Intrinsic feedback

- Feedback generated by the model itself
- No external information
- **Self-Refine** - Madaan et al., 2023
- **SelfCheck** - Miao et al., 2023

Extrinsic feedback

- Feedback from external sources/signals
- Retrieval, tools, rewards, evaluators
- **CRITIC** - Gou et al., 2023
- **Reflexion** - Shinn et al., 2023
- **Self-Correct** - Welleck et al., 2023

Our setting: ground truth is available for generating the hint, but the model must generate the corrective feedback itself.

Sufficiency Test

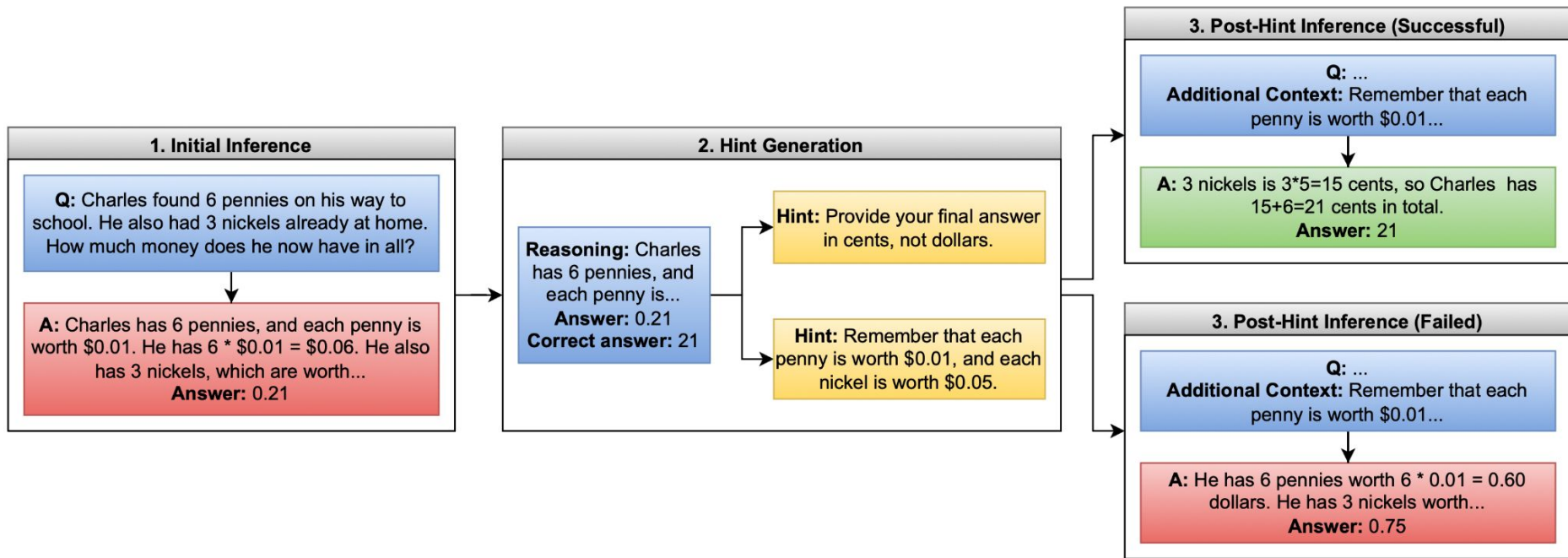


Fig.1. Overview of the three-step self-correction evaluation pipeline.

Measurement

We evaluate self-improvement as the **accuracy gain after hint injection**. For a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, let $\hat{y}_i^{(0)}$ be the initial prediction and $\hat{y}_i^{(h)}$ the post-hint prediction.

Initial Accuracy

$$\text{Acc}_{\text{init}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i^{(0)} = y_i]$$

Post-Hint Accuracy

$$\text{Acc}_{\text{hint}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i^{(h)} = y_i]$$

Accuracy Gain

$$\Delta\text{Acc} = \text{Acc}_{\text{hint}} - \text{Acc}_{\text{init}}$$

Higher ΔAcc indicates more successful self-correction.

Experimental Setup

- **Models:** 8 open-weight SLMs from 1.5B to 8B parameters: Llama-3.1-8B, Phi-4-mini, Qwen2.5-Math, Gemma-2-2B, and DeepSeek-R1 variants.
- **Benchmarks:** GSM8K, ASDiv, AQuA, AR-LSAT, and BIGBench Sports.
- **Token budgets:** 256, 512, 1024, 1536, and 2048 tokens.
- **Leakage control:** Hints revealing the answer are rejected up to 3 times; unresolved leakage is masked with <MSK>.
- **Sanity check:** Initial accuracies match the numbers reported by the model authors, validating our generation setup.

Results

Self-Correction Is Hard for SLMs

Accuracy before and after hint injection

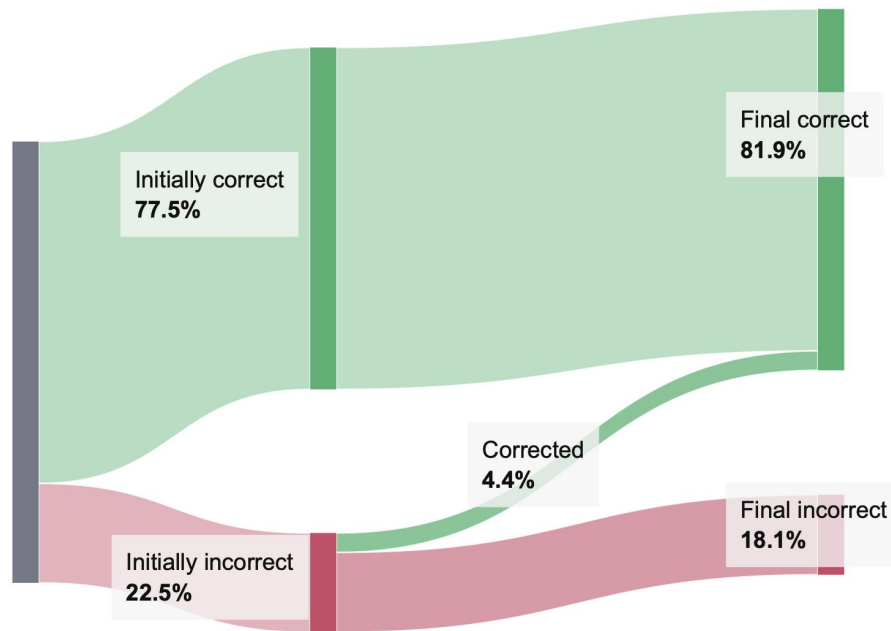


Fig. 2. Correction percentages after hint injection averaged across all of the outputs.

Key point: the bottleneck is not re-answering itself, but the model's inability to turn its own failed reasoning into useful feedback.

What Goes Wrong?

Too generic

“Check the reasoning carefully”

Format leaking

continues instructions or examples

Logically noisy

adds unrelated constraints

Repetitive

loops instead of diagnoses

Reasoning Models: No Clear Advantage

Reasoning vs Non-Reasoning Models (Averaged Across All Datasets)

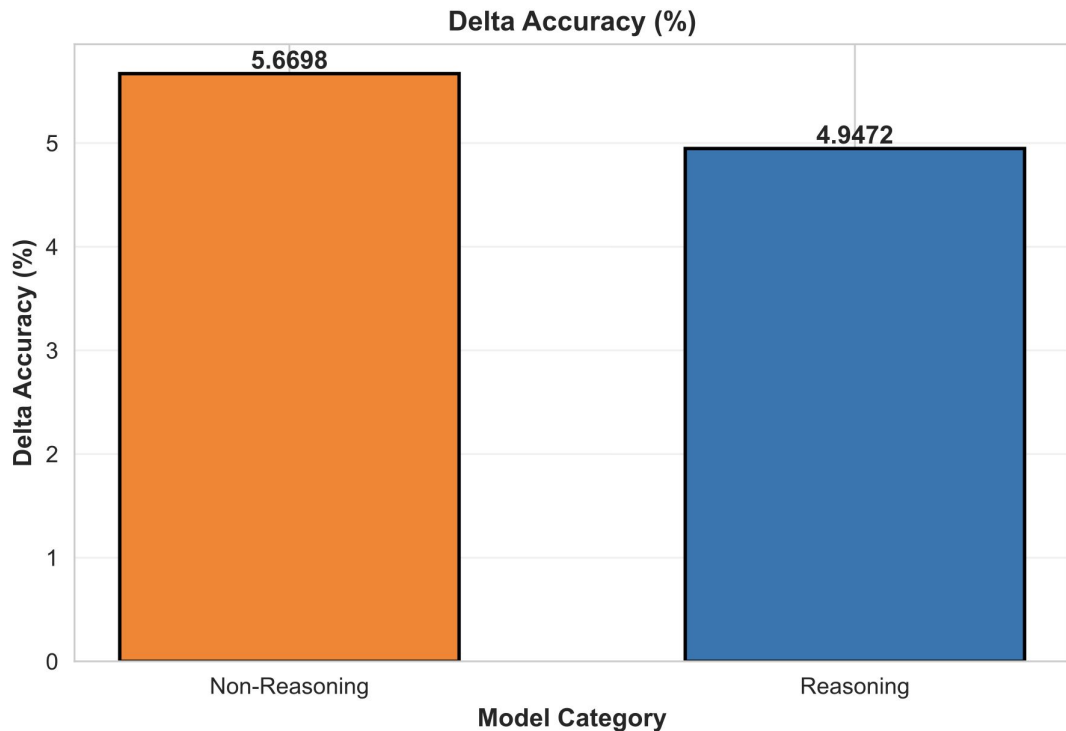


Fig. 3. Delta Accuracy of reasoning vs non-reasoning models averaged across all of the datasets.

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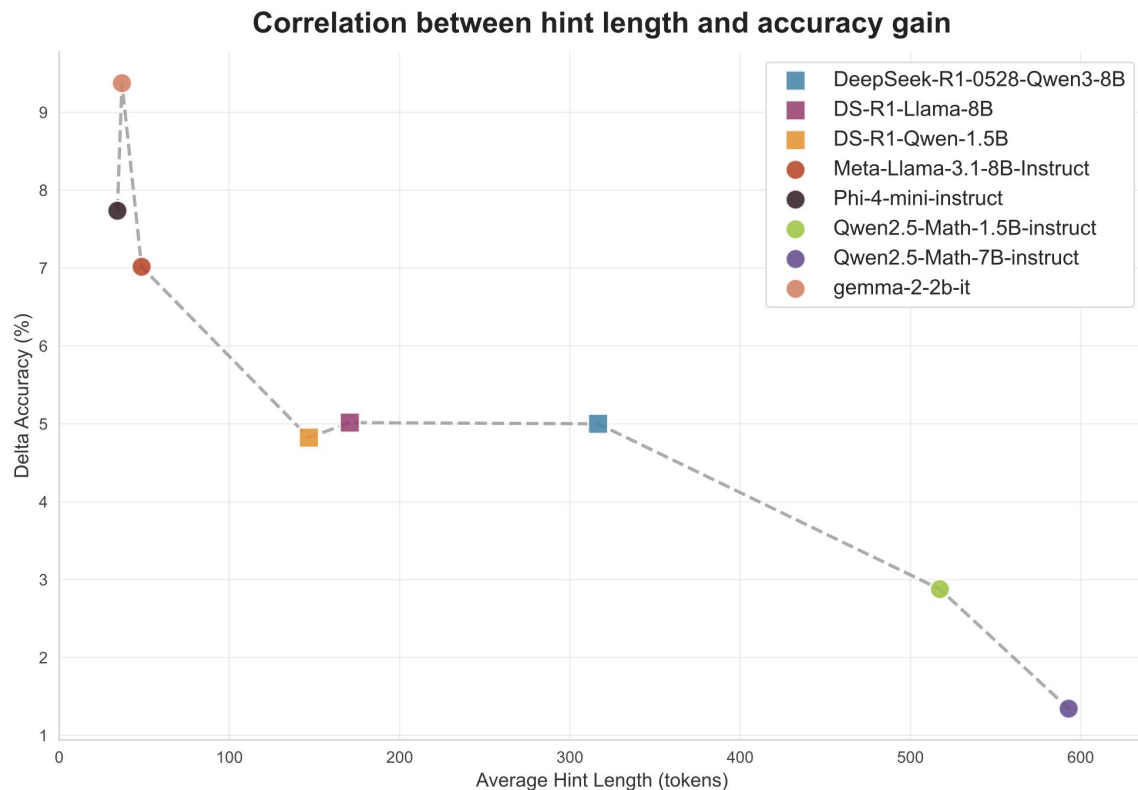


Fig. 4. Relationship between hint length and post-hint accuracy gain across models and datasets.

Useful vs. Useless Hints

Corrective and non-corrective hints have highly overlapping KeyNMF keywords.

Keyword	Corrected	Failed
model reasoning	16.1	15.4
total cost	16.1	12.8
basketball player	12.9	10.3
hockey player	12.9	15.4
answer boxed	9.7	12.8

Table 1. Semantic similarity of hint sentences that resulted in correct or incorrect answers.

Takeaways

I. Oracle access is not enough.

SLMs rarely turn ground truth into actionable self-feedback.

II. More text can hurt.

Long hints often drift into repetition, leakage, or irrelevant reasoning.

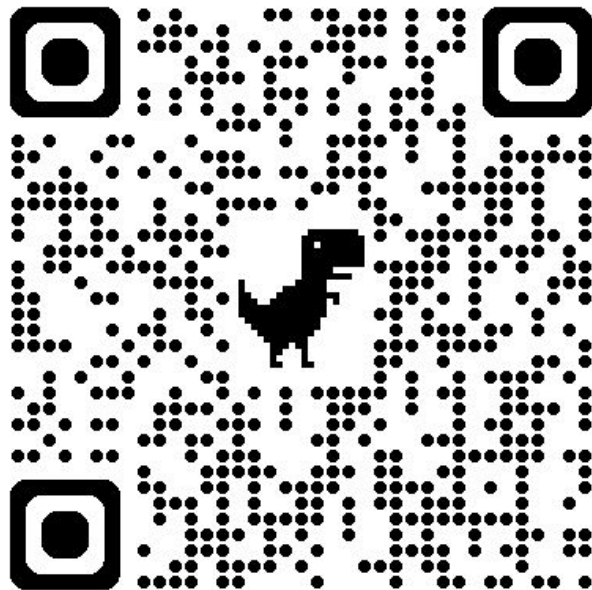
III. Self-correction needs better diagnostics.

The missing capability is recognizing the failed reasoning step, not merely sampling another answer.

Links



ACL Anthology



Codebase

Looking for a 2027 Internship

NLP · LLMs · Machine Learning

Interested in LLM reasoning, representation learning, and mathematically grounded ML.

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